

Probability density estimation

STA510

Rizzo chapters 12.1 and 12.2

Probability density estimation

- ▶ *Provided a sample $x_1, \dots, x_n$ from some unknown probability density function $f(x)$; how do we estimate $f(x)$?*
- ▶ Many applications:
 - ▶ useful for reporting/presenting results from statistical analysis or simulation.
 - ▶ checking/validating simulation code.
 - ▶ sometimes we need to estimate the value of the density $f(x)$ (or even derivatives, e.g. $f'(x)$) in some point x .
 - ▶ many more. . .

Two main avenues: histogram and KDE

- ▶ *histograms* (`hist()` in R)
- ▶ *kernel density estimates* (KDE) (`density()` (and others) in R)
- ▶ example: Given some data in a vector `x`:

```
length(x)
```

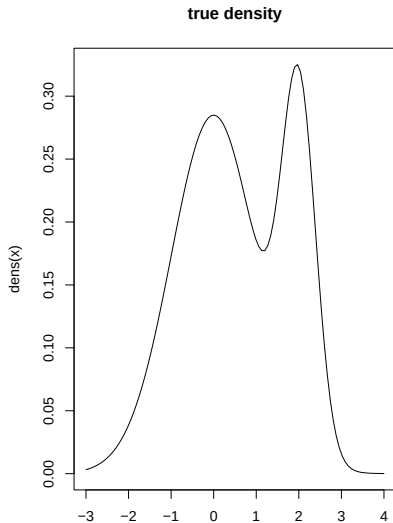
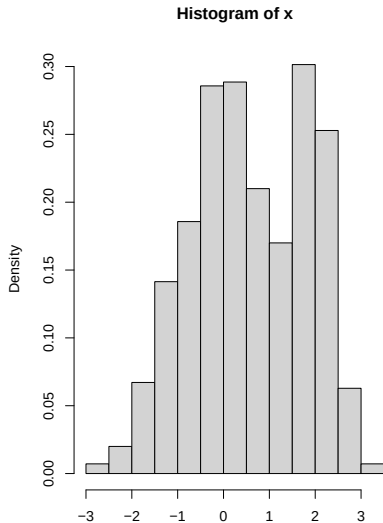
```
## [1] 1400
```

```
print(round(x[1:8],2))
```

```
## [1] -0.56 -0.23  1.56  0.07  0.13  1.72  0.46 -1.27
```

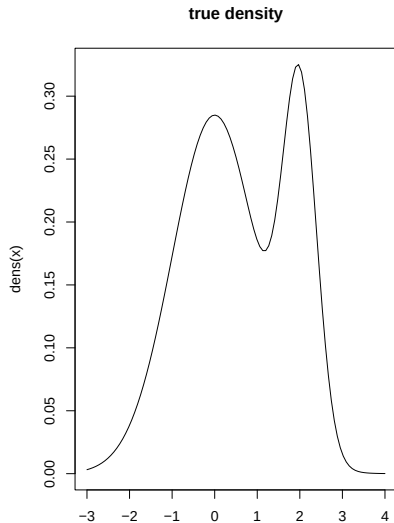
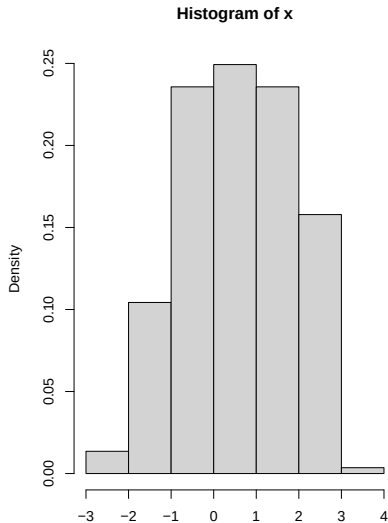
Intro example: Histogram (default)

```
par(mfrow=c(1,2))  
hist(x,probability=TRUE)  
curve(dens,from=-3,to=4,main="true density")
```



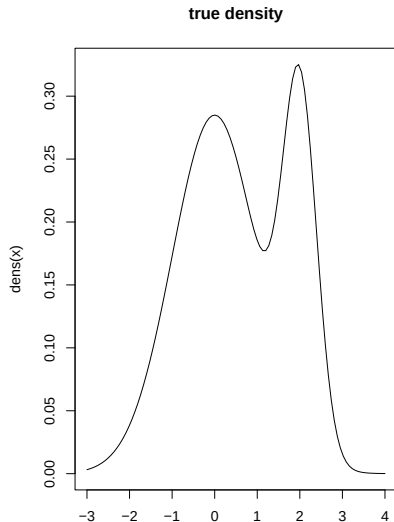
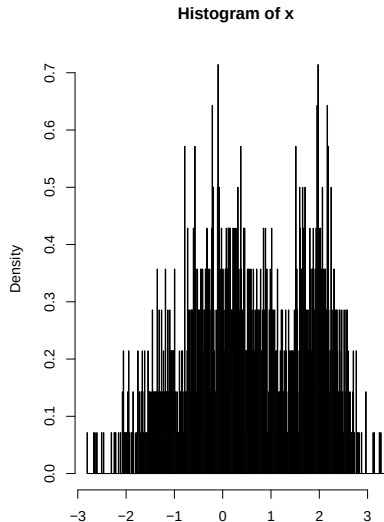
Intro example: Histogram (too few bins)

```
par(mfrow=c(1,2))  
hist(x,breaks=5,probability=TRUE)  
curve(dens,from=-3,to=4,main="true density")
```



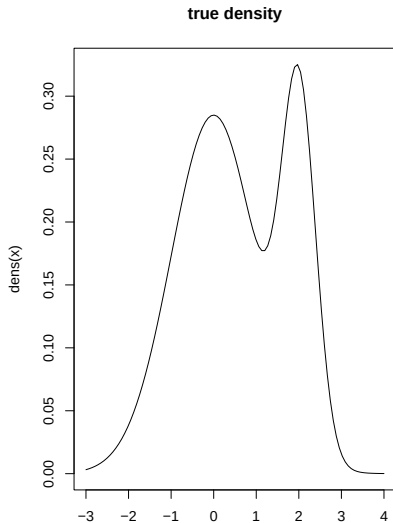
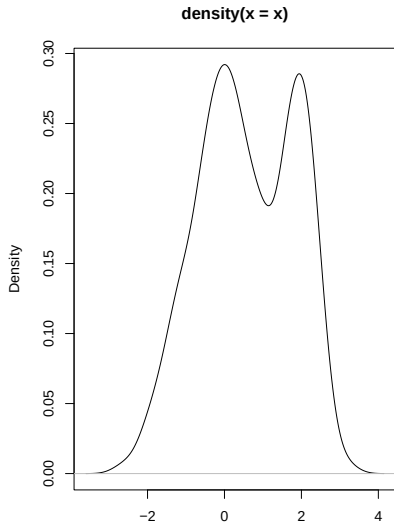
Intro example: Histogram (too many bins)

```
par(mfrow=c(1,2))  
hist(x,breaks=500,probability=TRUE)  
curve(dens,from=-3,to=4,main="true density")
```



Intro example: KDE (default)

```
par(mfrow=c(1,2))  
plot(density(x))  
curve(dens,from=-3,to=4,main="true density")
```



Intro example: KDE (default)

- ▶ let's look more carefully at what `density()` returns:

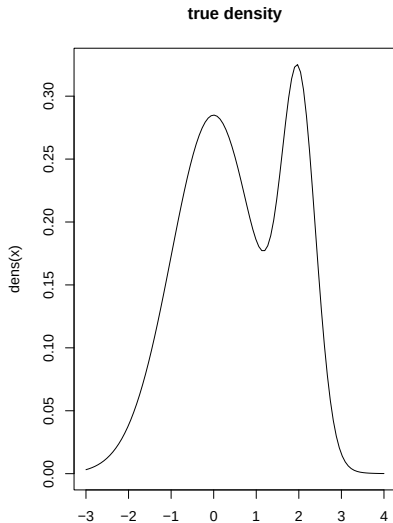
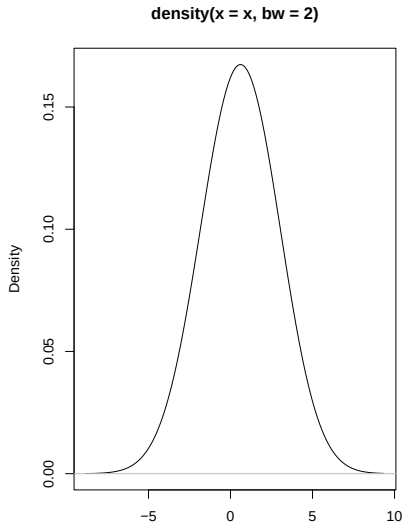
```
density(x)
```

```
##  
## Call:  
## density.default(x = x)  
##  
## Data: x (1400 obs.); Bandwidth 'bw' = 0.2635  
##  
##           x                y  
## Min.      :-3.6003    Min.    :1.669e-05  
## 1st Qu.   :-1.6636    1st Qu.:1.050e-02  
## Median    : 0.2732    Median :1.264e-01  
## Mean      : 0.2732    Mean    :1.288e-01  
## 3rd Qu.   : 2.2099    3rd Qu.:2.321e-01  
## Max.      : 4.1467    Max.    :2.921e-01
```

- ▶ Note: “bandwidth”

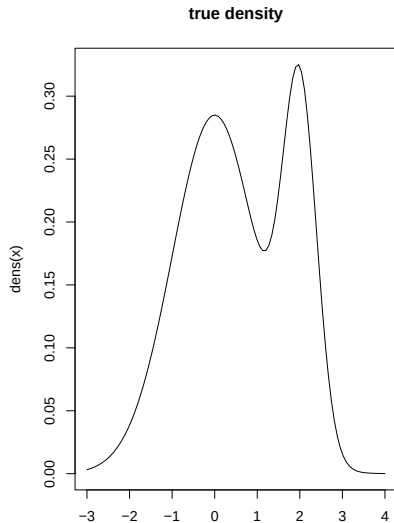
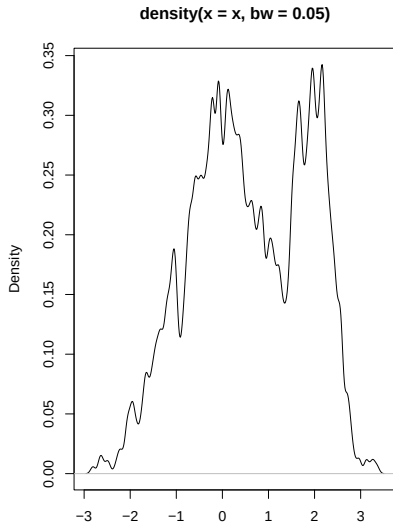
Intro example: KDE (too much smoothing)

```
par(mfrow=c(1,2))  
plot(density(x,bw=2.0))  
curve(dens,from=-3,to=4,main="true density")
```



Intro example: KDE (too little smoothing)

```
par(mfrow=c(1,2))  
plot(density(x,bw=0.05))  
curve(dens,from=-3,to=4,main="true density")
```



Objectives here

- ▶ Understand the basics of how these two methods work, and thereby their limitations.
- ▶ The nature of smoothing (number of bins in hist, bandwidth in KDE)
- ▶ Understand the “bias-variance tradeoff”.
- ▶ NOT: all the technical details not covered in these notes.

Histograms

Histogram (basics)

- ▶ As before; we have a sample $x_1, \dots, x_n$ from some unknown density $f(x)$
- ▶ Consider a fixed uniform grid $t_1 < t_2 < \dots < t_{K+1}$ on the x-axis covering all our observations.
- ▶ Uniform: $t_{k+1} - t_k = h$ for each $k = 1, \dots, K$
- ▶ Then our estimate of the density $f(x)$ is given by

$$\hat{f}(x) = \frac{\nu_k}{nh} \text{ when } t_k \leq x < t_{k+1}$$

where ν_k is the number of x_i s in the interval $[t_k, t_{k+1})$

- ▶ In R: `hist()` with extra argument: `probability=TRUE`

Histogram (theoretical basics)

- Note: histogram is a proper density!

$$\int \hat{f}(x) dx = \sum_k \int_{t_k}^{t_{k+1}} \frac{\nu_k}{nh} dx = \sum_k \frac{\nu_k}{nh} h = \frac{1}{n} \underbrace{\sum_k \nu_k}_{=n} = 1$$

- For some fixed x , $\hat{f}(x)$ is an estimator of $f(x)$ that is both *biased* and has *non-zero variance*:
- suppose $t_k \leq x < t_{k+1}$, then

$$\nu_k \sim \text{binomial}(n, p_k), \quad p_k = \int_{t_k}^{t_{k+1}} f(y) dy$$

- Recall $E(X) = np$, $\text{Var}(X) = np(1-p)$ if $X \sim \text{binomial}(n, p)$

Histogram (bias)

- Bias of $\hat{f}(x)$:

$$E(\hat{f}(x)) = E\left(\frac{\nu_k}{nh}\right) = \frac{1}{nh}E(\nu_k) = \frac{1}{nh}np_k = \frac{p_k}{h},$$

but

$$\frac{p_k}{h} = \underbrace{\frac{1}{h} \int_{t_k}^{t_k+h} f(y) dy}_{\text{mean value of } f(x)} \neq f(x)$$

unless $f(x)$ is constant! Therefore is $\hat{f}(x)$ a biased estimator of $f(x)$.

- However; choosing the grid so that $x = t_k$, then $\lim_{h \rightarrow 0} \frac{1}{h} \int_{t_k}^{t_k+h} f(y) dy = f(t_k) = f(x)$ (if $f(x)$ is smooth), hence we could in principle remove the bias by letting $h \rightarrow 0$ (but this would give us infinitely many bins. . .)
- TAKEAWAY: *bias* grows as h grows (fewer bins, more bias, ref figure above with too few bins).

Histogram (variance)

-Variance of $\hat{f}(x)$:

$$\begin{aligned}\text{Var}(\hat{f}(x)) &= \text{Var}\left(\frac{\nu_k}{nh}\right) = \frac{1}{(nh)^2} \text{Var}(\nu_k) = \frac{np_k(1-p_k)}{n^2 h^2} \\ &= \frac{p_k}{nh^2} - \frac{p_k^2}{nh^2} \approx \frac{f(x)}{nh} - \frac{f(x)^2}{n}\end{aligned}$$

where the last approximation hold for “small” h (recall $\lim_{h \rightarrow 0} \frac{1}{h} \int_{t_k}^{t_k+h} f(x) dx = f(x)$)

- ▶ To summarize; for “small” h (and large n), then $\text{Var}(\hat{f}(x)) \approx \frac{f(x)}{nh}$.
- ▶ TAKEAWAY: Removing bias by letting $h \rightarrow 0$ would make the *variance* explode!
- ▶ SIDENOTE: by choosing h sufficiently large, we get a single bin with height that has zero variance.

bias-variance tradeoff

- ▶ Two takeaways: we should choose h *small enough so that the bias is not too large*, but we should choose h *large enough so that the variance is not too large*...
- ▶ One approach is based on minimizing the mean-squared error (MSE) (for fixed x):

$$MSE(\hat{f}(x)) = E(\hat{f}(x) - f(x))^2 = \text{Var}(\hat{f}(x)) + \underbrace{[E(\hat{f}(x) - f(x))]^2}_{= \text{squared bias}}$$

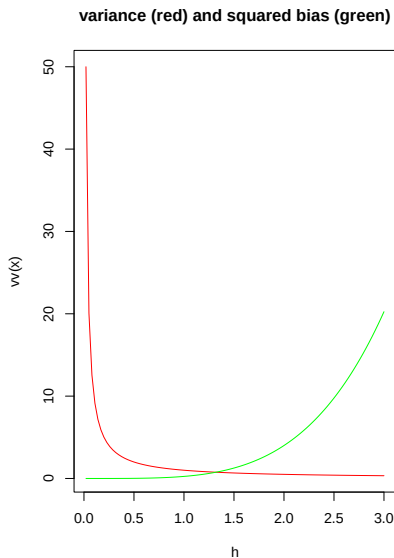
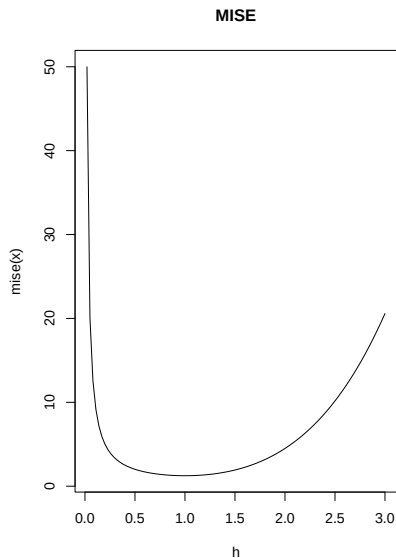
or more generally mean-integrated squared error (MISE):

$$MISE(\hat{f}) = \int MSE(\hat{f}(x)) dx$$

- ▶ Always a tradeoff between variance term and squared bias term.

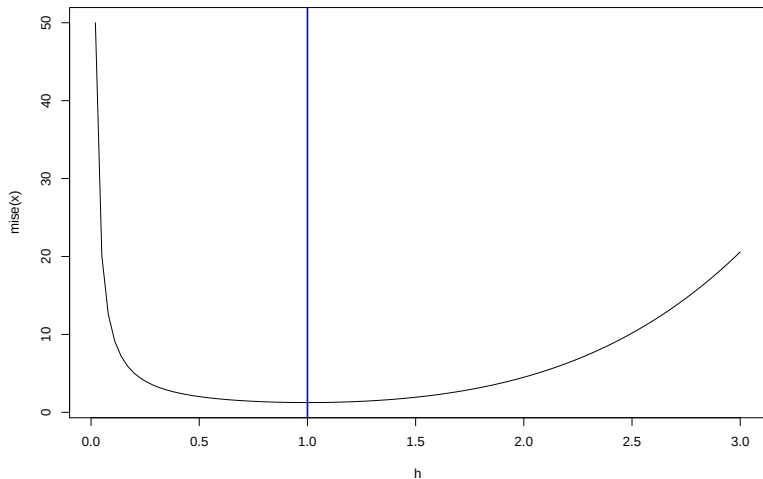
bias-variance tradeoff

- MSE or MISE as function of h typically looks something like this:



choose a tradeoff

- We choose a value of h where the MISE is minimized:



Rules for choosing h

- ▶ Many rules exist for choosing h , depending on assumptions made.
- ▶ For the histogram, it can be shown that (for small h and large n)

$$MISE = \frac{1}{nh} + \frac{h^2}{12} \int f'(x)^2 dx + O\left(\frac{1}{n} + h^3\right)$$

- ▶ Minimizing the leading terms of this expression as a function of h yields

$$h_n^* = \left(\frac{6}{n \int f'(x)^2 dx} \right)^{1/3}$$

- ▶ So far not operational as it depends on unknown quantity $\int f'(x)^2 dx$ - need to make further assumptions

Rules for choosing h - normal assumption

- ▶ Assuming data $x_1, \dots, x_n$ are normal, then

$$\int f'(x)^2 dx = \int_{-\infty}^{\infty} \left[\frac{d}{dx} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-(x - \mu)^2/(2\sigma^2)) \right]^2 dx \\ \approx 0.1410473959\sigma^{-3}$$

- ▶ Yields Scott's Normal Reference rule:

$$\hat{h} = (6/0.1410473959)^{1/3}(\sigma^3)^{1/3}n^{-1/3} \approx 3.49\hat{\sigma}n^{-1/3}$$

where $\hat{\sigma}$ is an estimator of the standard deviation of the data (typically s).

Sturges' Rule

- ▶ Very common.
- ▶ Based on more heuristic reasoning; yields

$$\hat{h} = \frac{\max_i x_i - \min_i x_i}{1 + \log_2(n)}$$

where $\log_2$ is the base 2 logarithm (`log2()` in R).

example - Scott's Normal Reference rule:

- Note: R rounds h to a “nice” number...

```
h.out <- hist(x,breaks="Scott",plot=FALSE)
print(paste0("h used by R : ",
             h.out$breaks[2]-h.out$breaks[1]))
```

```
## [1] "h used by R : 0.5"
```

```
print(paste0("h suggested by Scott's rule : ",
             3.49*sd(x)*length(x)^(-1/3) ))
```

```
## [1] "h suggested by Scott's rule : 0.388943633178186"
```

example - Sturges' Rule:

- Note: R rounds h to a “nice” number...

```
h.out <- hist(x,plot=FALSE) #default method in R  
print(paste0("h used by R : ",  
             h.out$breaks[2]-h.out$breaks[1]))
```

```
## [1] "h used by R : 0.5"
```

```
print(paste0("h suggested by Sturges' Rule : ",  
             (max(x)-min(x))/(1+log2(length(x)))))
```

```
## [1] "h suggested by Sturges' Rule : 0.538451605446172"
```

Kernel density estimation (KDE)

Basics of KDE

- ▶ Let $K(\cdot)$ be some symmetric probability density function with mean=0, variance=1, e.g. standard normal pdf.
- ▶ I.e. $\int K(x)dx = 1$, $\int xK(x)dx = 0$, $\int x^2K(x)dx = 1$.
- ▶ The KDE for some independent sample $X_1, \dots, X_n$ is then given by

$$\hat{f}_K(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - X_i}{h}\right)$$

$h > 0$ is the “bandwidth” (smoothing parameter).

- ▶ Much more flexible than histograms, but more things can also go wrong... (both have their pros and cons)
- ▶ Note: $\int \hat{f}_K(x)dx = 1$ i.e. proper density.

Example

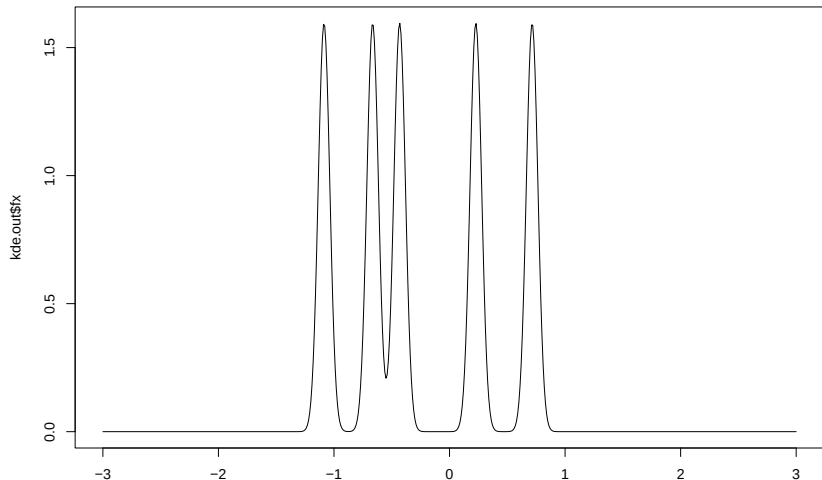
- ▶ “Lazy” implementation based on Gaussian kernel:

```
my.kde <- function(x,X,h){ # allow for a vector of x-s  
  fx <- numeric(length(x)) # allocate space  
  for(i in 1:length(x)){  
    #eqn in previous slide  
    fx[i] <- mean(dnorm(x[i],mean=X,sd=h))  
  }  
  return(list(x=x,fx=fx,h=h))  
}
```

- ▶ Note: if K is the $N(0,1)$ -pdf, then $h^{-1}K(h^{-1}(x - X_i))$ is the $N(X_i, h^2)$ -pdf evaluated at x .

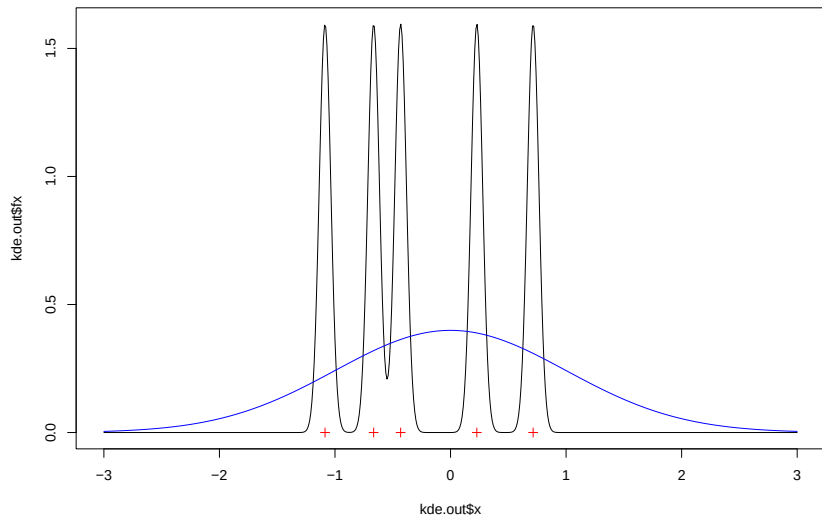
Example continued

```
X <- rnorm(5) # tiny sample for illustration  
x.grid <- seq(from=-3.0,to=3.0,by=0.01)  
kde.out <- my.kde(x=x.grid,X=X,h=0.05)  
plot(kde.out$x,kde.out$fx,type="l")
```



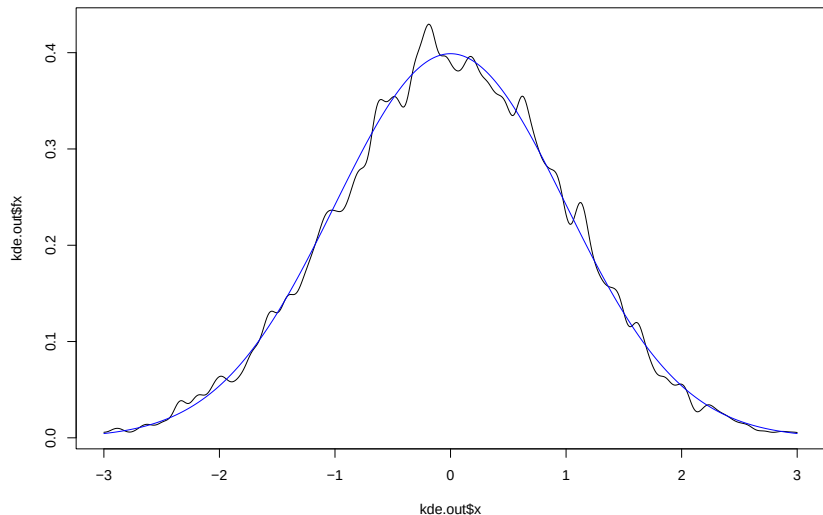
Example continued

```
plot(kde.out$x,kde.out$fx,type="l")  
points(X,0*X,pch=3,col="red")  
lines(x.grid,dnorm(x.grid),col="blue")
```



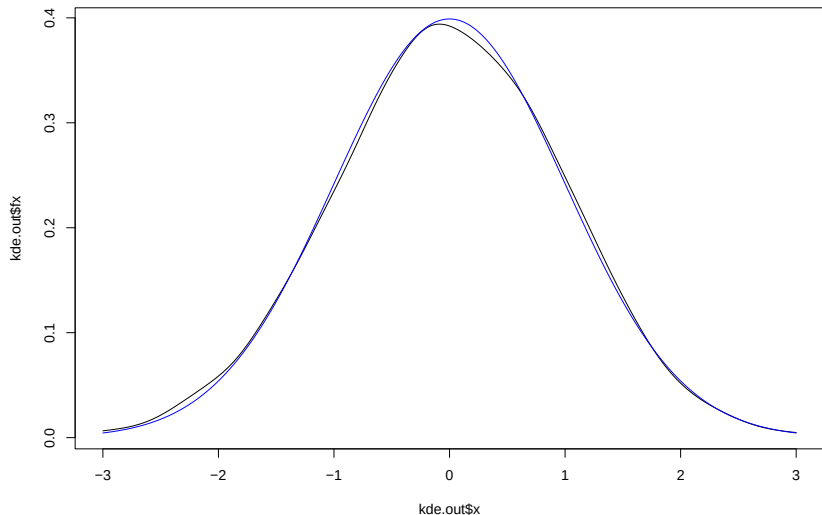
Example continued, More data

```
X <- rnorm(5000); kde.out <- my.kde(x=x.grid,X=X,h=0.05)
plot(kde.out$x,kde.out$fx,type="l")
lines(x.grid,dnorm(x.grid),col="blue")
```



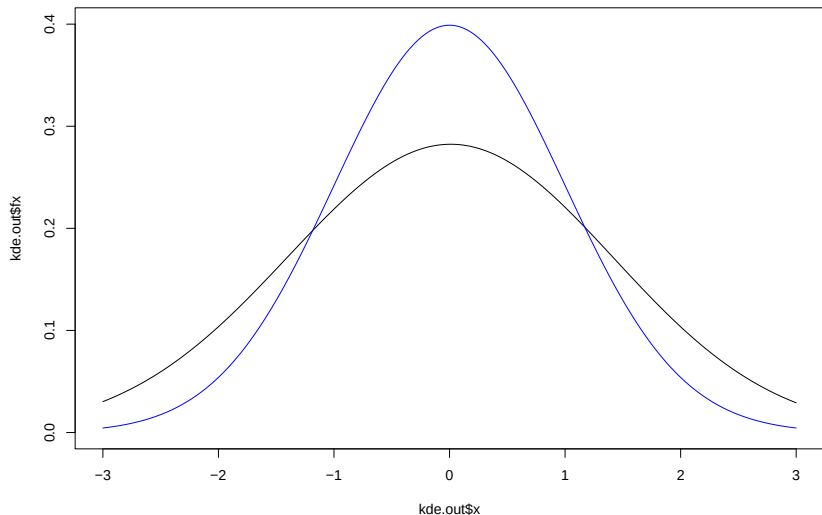
Example continued, More smoothing

```
kde.out <- my.kde(x=x.grid,X=X,h=0.2)  
plot(kde.out$x,kde.out$fx,type="l")  
lines(x.grid,dnorm(x.grid),col="blue")
```



Example continued, Too much smoothing

```
kde.out <- my.kde(x=x.grid,X=X,h=1)
plot(kde.out$x,kde.out$fx,type="l",ylim=c(0,0.4))
lines(x.grid,dnorm(x.grid),col="blue")
```



Choosing the bandwidth

- ▶ The bandwidth parameter h plays very much the same role as the bin-width in the histogram.
- ▶ Generally:
 - ▶ Small h leads to low bias but high variance.
 - ▶ Large h leads to low variance and high bias.
- ▶ Note:

$$E(\hat{f}_K(x)) = \int h^{-1}K(h^{-1}(x-y))f(y)dy = f(x) + \underbrace{\frac{h^2 f''(x)}{2}}_{\text{bias}} + O(h^4)$$

for small $h > 0$. For small h and large n , $\text{Var}(\hat{f}_K(x)) \propto (nh)^{-1}$.

MISE for the KDE:

- ▶ Still assuming $\int x^2 K(x) dx = 1$, then the (AMISE) optimal bandwidth is

$$\hat{h} = \left[\frac{\int (K(x))^2 dx}{\int (f''(x))^2 dx} \right]^{1/5} n^{-1/5}$$

Again not directly applicable since it depends on the unknown $f''(x)$.

- ▶ For standard normal K and $f(x)$ corresponding to a $N(\mu, \sigma^2)$ -distribution;

$$\int (K(x))^2 dx = \frac{1}{2\sqrt{\pi}}, \quad \int (f''(x))^2 dx \approx 0.2115710938 \sigma^{-5}$$

we get “the rule of thumb”: $\hat{h} \approx 1.06 \hat{\sigma} n^{-1/5}$ where $\hat{\sigma}$ is an estimate of σ .

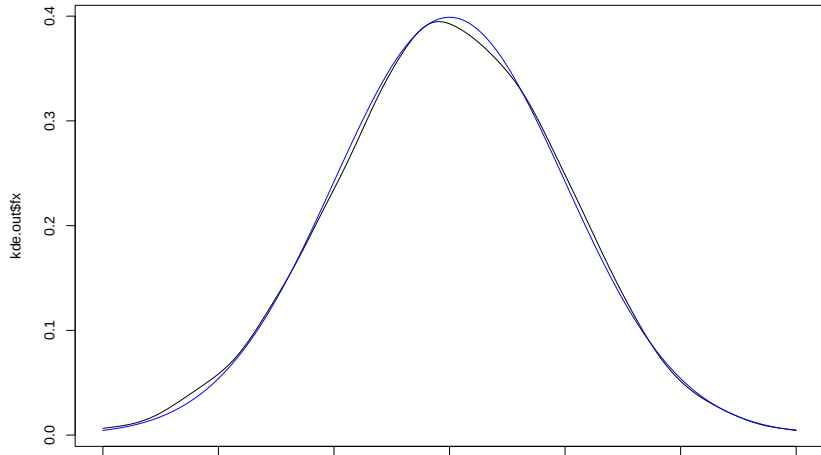
Back to example; function redefinition

```
my.kde <- function(x,X,h=1.06*sd(X)*length(X)^(-0.2)){  
  fx <- numeric(length(x)) # allocate space  
  for(i in 1:length(x)){  
    fx[i] <- mean(dnorm(x[i],mean=X,sd=h))  
  }  
  return(list(x=x,fx=fx,h=h))  
}
```

Try new function

```
kde.out <- my.kde(x=x.grid,X=X)
plot(kde.out$x,kde.out$fx,type="l",
     main=paste0("bandwidth = ",kde.out$h))
lines(x.grid,dnorm(x.grid),col="blue")
```

bandwidth = 0.192902742851946

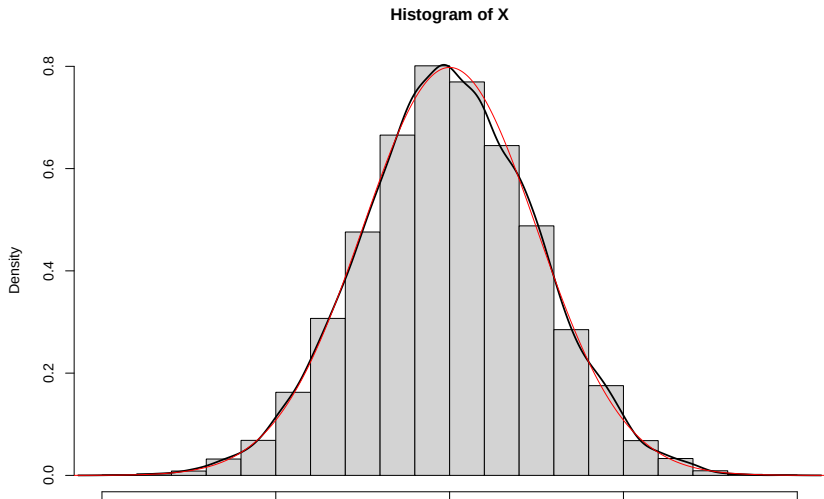


KDE in practice

- ▶ In practice, one typically uses the built in `density()`-function for KDEs.
- ▶ `density()` allows for different kernels K and has automatic bandwidth selection.
- ▶ BUT RECALL: **KDEs are (generally biased) estimators of the density $f(x)$, not the density itself.**
- ▶ Bad things can happen if the data deviates much from normality (which the histogram is more robust against).

Examples: Normal

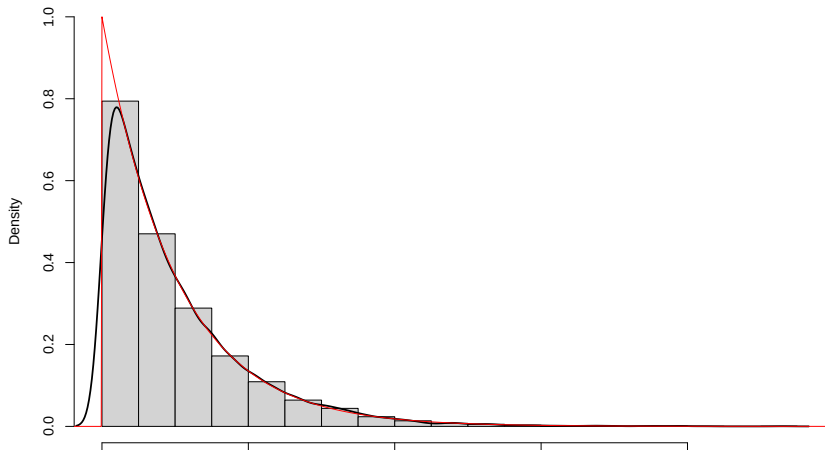
```
X <- rnorm(10000,mean=2,sd=0.5)
hist(X,probability=TRUE);lines(density(X),lwd=2)
xg <- seq(from=-1,to=5,by=0.01)
lines(xg,dnorm(xg,mean=2,sd=0.5),col="red")
```



Examples: Exponential (KDE fails around $x = 0$)

```
X <- rexp(10000)
hist(X,probability=TRUE,ylim=c(0,1));lines(density(X),lwd=2)
xg <- seq(from=-1,to=15,by=0.01)
lines(xg,dexp(xg),col="red")
```

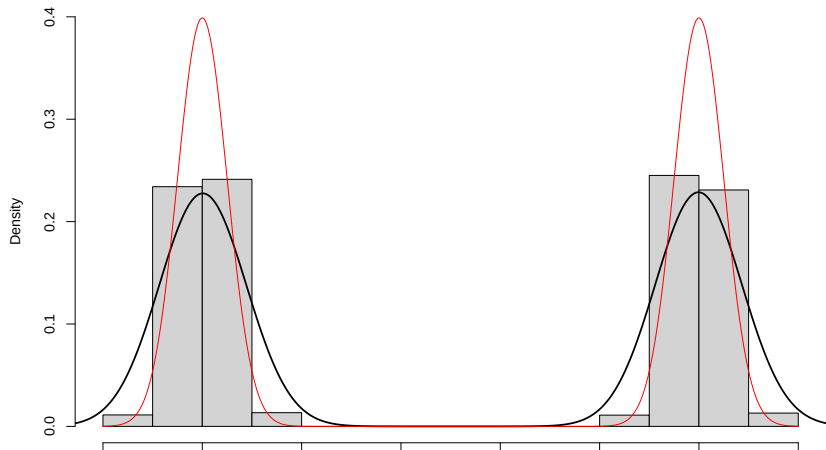
Histogram of X



Examples: Normal mixture (both fail really...)

```
X <- c(rnorm(5000,sd=0.5),rnorm(5000,mean=10,sd=0.5))  
hist(X,probability=TRUE,ylim=c(0,0.4));lines(density(X),lwd  
xg <- seq(from=-2,to=12,by=0.01)  
lines(xg,0.5*dnorm(xg,sd=0.5)+0.5*dnorm(xg,mean=10,sd=0.5))
```

Histogram of X



Histogram, KDE or ECDF?

- ▶ Histogram and KDE estimate the unknown density $f(x)$.
 - ▶ Useful for visualizing the shape of a distribution, e.g. modes, skewness and tails.
 - ▶ Both require a smoothing choice: Bin-width for the histogram; Bandwidth for the KDE.

- ▶ The ECDF

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$$

estimates the distribution function $F(x)$ directly.

- ▶ An important advantage of the ECDF is that it has **no bin-width or bandwidth parameter to choose**.

Histogram, KDE or ECDF?

- ▶ Thus, if the CDF contains the information we need, the ECDF is often a better choice. R-code: `HistKDE.R`
- ▶ However, the ECDF and density estimators answer different questions:
 - ▶ ECDF: probabilities and quantiles.
 - ▶ Histogram/KDE: local shape and density.

What should we take away?

- ▶ Histogram and KDE are useful nonparametric estimators of a density $f(x)$, but both involve a smoothing choice.
- ▶ The amount of smoothing gives a **bias–variance tradeoff**:
 - ▶ too little smoothing gives a noisy estimate,
 - ▶ too much smoothing hides features of the distribution.
- ▶ If estimation of the CDF is sufficient, the ECDF avoids the choice of a smoothing parameter altogether.
- ▶ Large literature on e.g. different choices of K , higher dimensions, and so on.

What should we take away?

- ▶ The estimators discussed here assume that the data live on the real line. The **support and geometry of the data matter**.
- ▶ If additional information is available, it should be incorporated into the estimator. For example:
 - ▶ $X > 0$: use boundary kernels (page 359);
 - ▶ periodic/directional data: 0° and 360° represent the same direction, so ordinary KDE is inappropriate near the boundary.
- ▶ For directional data on the circle, a **circular KDE** can instead be used. See R-code: SolaWindDirections.R